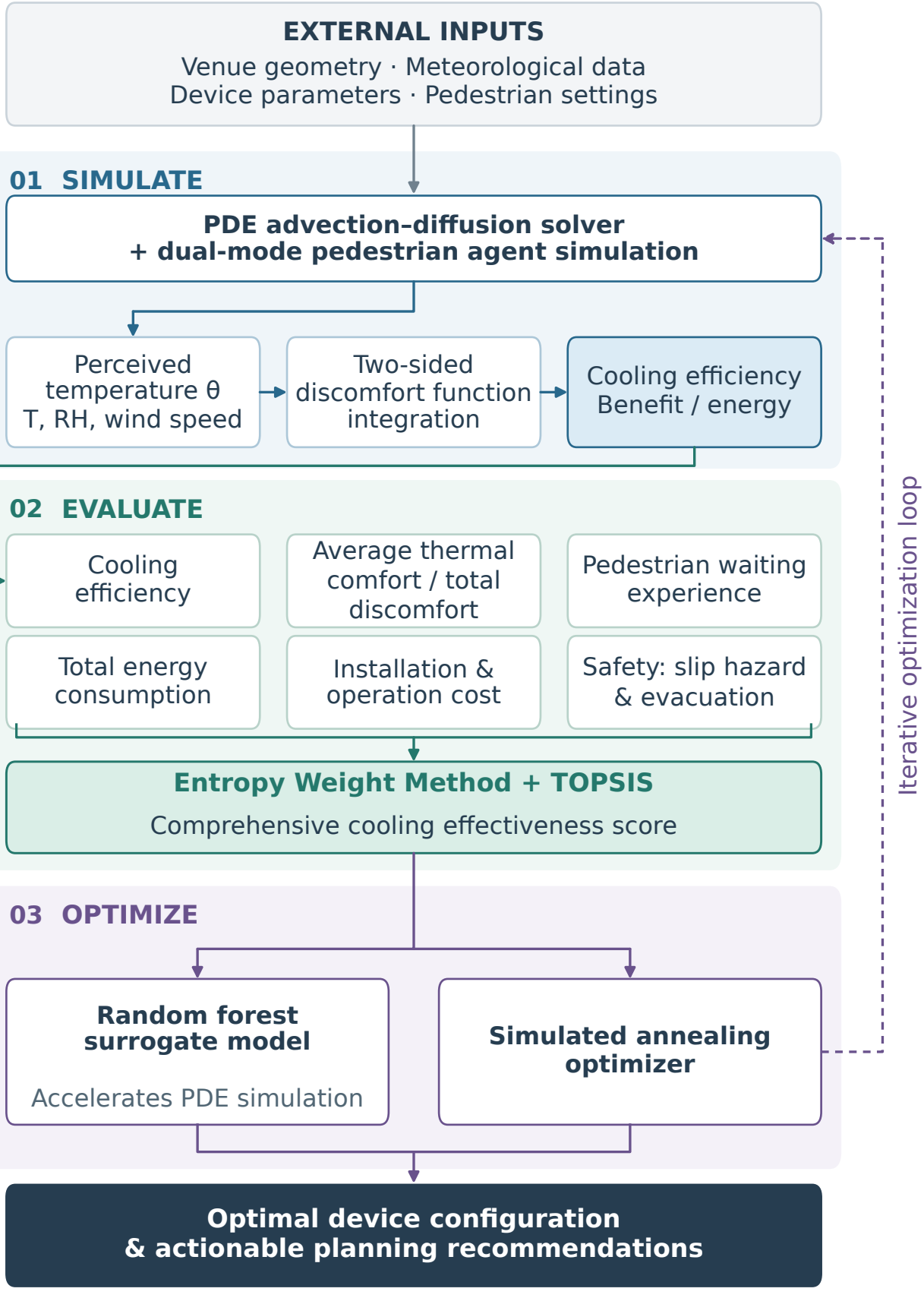
EXTERNAL INPUTS
Venue geometry · Meteorological data
Device parameters · Pedestrian settings
01 SIMULATE
PDE advection–diffusion solver
+ dual-mode pedestrian agent simulation
Perceived temperature θ
T, RH, wind speed
Two-sided discomfort function integration
Cooling efficiency
Benefit / energy
02 EVALUATE
Cooling efficiency
Average thermal comfort / total discomfort
Pedestrian waiting experience
Total energy consumption
Installation & operation cost
Safety: slip hazard & evacuation
Entropy Weight Method + TOPSIS
Comprehensive cooling effectiveness score
03 OPTIMIZE
Random forest surrogate model
Accelerates PDE simulation
Simulated annealing optimizer
Optimal device configuration
& actionable planning recommendations
Iterative optimization loop